

Quiz 5 – Applied OS: Cyber-Defense & Agentic AI

Student Name: _____

Q1	Q2	Total
(4 + 2)	(4)	(10)

- Q1 A system is running the Linux Completely Fair Scheduler (CFS). Task P has a priority weight ($W_P = 1024$), (4)
a) and Task Q has a high priority ($W_Q = 4096$). Both tasks are given a physical CPU timeslice of 12ms.

Calculate the **vruntime** increase for both tasks.

Solution:

- Task P Calculation:

$$vruntime\ increase = actual\ time \times \left(\frac{W_0}{W_P} \right) = 12ms \times \left(\frac{1024}{1024} \right) = 12ms \times 1 = 12ms$$

- Task Q Calculation

$$vruntime\ increase = actual\ time \times \left(\frac{W_0}{W_Q} \right) = 12ms \times \left(\frac{1024}{4096} \right) = 12ms \times \frac{1}{4} = 3.0\ ms$$

- b) (2)

According to the Red-Black Tree mechanics described in the slides, explain how this mathematical manipulation of time ensures Task Q gets more CPU time in the long run.

- The CFS rule states that the scheduler always picks the leftmost node in the Red-Black Tree, which contains the task with the smallest vruntime.
- Even though both tasks ran for 12ms physically, Task Q's vruntime grew 4 times slower.
- Because its vruntime stays smaller for a longer period, it will remain on the left side of the tree and be continuously selected by the scheduler, resulting in more physical CPU time overall.

Q2 An operating system uses Multi-queue Multiprocessor Scheduling (MQMS) on a dual-core system (CPU 0 and CPU 1). Four compute-heavy, equal-priority tasks (A, B, C, and D) enter the system. The scheduler places tasks A, B, and C into Queue 0 (assigned to CPU 0) and places task D into Queue 1 (assigned to CPU 1). Both queues implement a Round Robin scheduling discipline.

a) Using a diagram, show how the 4 tasks will be executed (with load imbalance).

(2)

CPU_0	A	B	C	A	B	C	A	B	C	A	B	C	...
CPU_1	D	D	D	D	D	D	D	D	D	D	D	D	...

b) Using a diagram, show how the 4 tasks will be executed with load balancing.

(2)

CPU_0	A	B	A	B	A	B	A	B	A	B	A	B	...
CPU_1	C	D	C	D	C	C	D	C	D	C	D	C	...